MATHEMATICS

Analysis and approaches

Higher Level | Set A | Paper 3

Mock examination · 2025 series

Time allowed: 1 hour 15 minutes Maximum mark: 55

Candidate name: _____

Candidate number: _____

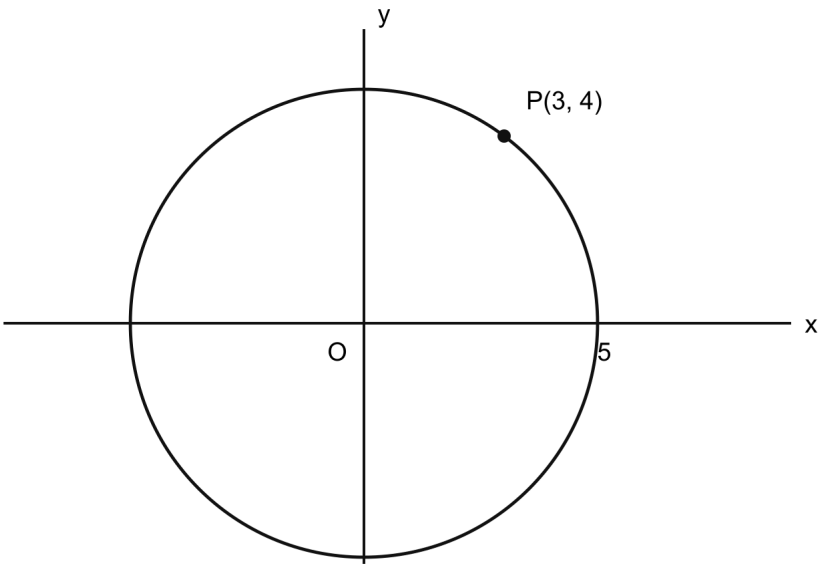
Date: _____

Instructions

- A graphic display calculator is required.
- This paper contains two compulsory extended problem-solving questions.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

Question 1 [Maximum mark: 25]

The circle C has equation $x^2 + y^2 = 25$. The point $P = (3, 4)$ lies on its upper semicircle, written $y = f(x)$. Coordinates are measured in millimetres. In parts(b)to(e), consider the differential equation $\mathrm{d}y/\mathrm{d}x = -x/y$, with $y = 4$ when $x = 3$, taking the positive branch near this point.



Circle C; coordinates measured in millimetres

(a)(i)) Use implicit differentiation to show that $\mathrm{d}y/\mathrm{d}x = -x/y$ on C where $y \neq 0$. [2]

(a)(ii)) Find the tangent to C at P. [2]

(b)(i)) State, with a reason, whether the differential equation is separable. [1]

(b)(ii)) State, with a reason, whether it is homogeneous in the sense $y\prime = F(y/x)$. [1]

(b)(iii)) State, with a reason, whether it is a first-order linear equation in y. [1]

(c) Solve the initial-value differential equation to obtain an implicit equation relating x and y. [3]

(d)(i)) Use Euler's method with step size 0.1, starting at $(3, 4)$, to estimate y when $x = 4$. [2]

(d)(ii)) Find the exact value of y at $x = 4$ and explain why the Euler estimate is an overestimate. [2]

(e)(i)) Show that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{x^2 + y^2}{y^3}$. [3]

(e)(ii)) Find the Maclaurin approximation to $f(x)$ through the term in x^2. [3]

(e)(iii)) Use this approximation to estimate $f(4)$. [1]

(f) A particle moves on C. At the instant it passes through P, its x-coordinate is increasing at 8 mm s^{-1}. Find the rate of change of its y-coordinate at that instant. [4]

Question 2 [Maximum mark: 30]

For an integer $n \geq 2$, consider the equation $z^n = 1$. Write $\operatorname{cis}\theta = \cos\theta + i\sin\theta$ and define $z_k = \operatorname{cis}(2\pi k/n)$ for integer k. Any n consecutive indices give all distinct roots. Let S_n and P_n be respectively the sum and product of these n distinct roots. You may use $S_n = 0$ for $n \geq 2$.

(a) Verify that every z_k satisfies $z^n = 1$. [1]

(b) State a formula for P_n in terms of n. [1]

(c)(i)) For $n = 3$, write z_{-1}, z_0, z_1 in Cartesian form. [2]

(c)(ii)) Use these roots to calculate P_3. [2]

(d)(i)) For $n = 4$, write z_{-1}, z_0, z_1, z_2 in Cartesian form. [2]

(d)(ii)) Use these roots to calculate P_4. [1]

(e) Show that $(z - z_k)(z - z_{-k}) = z^2 - 2z\cos(2\pi k/n) + 1$. [3]

(f)(i)) For $n = 6$, list all six roots in cis form using the indices $-2, -1, 0, 1, 2, 3$. [3]

(f)(ii)) Write z_3 in Cartesian form for $n = 6$. [1]

(f)(iii)) Factorize $z^6 - 1$ as a product of real linear and real quadratic factors. [2]

(g)(i)) For $n = 5$, list all five roots in cis form using indices $-2, -1, 0, 1, 2$. [2]

(g)(ii)) Factorize $z^5 - 1$ into one real linear and two real quadratic factors, leaving cosine coefficients exact. [2]

(h) Write down the analogous real factorization of $z^7 - 1$. [2]

(i) Using $S_5 = 0$, prove that $\cos(2\pi/5) + \cos(4\pi/5) = -1/2$. [4]

(j)(i)) Write down $\cos(2\pi/7) + \cos(4\pi/7) + \cos(6\pi/7)$. [1]

(j)(ii)) For $m \in \mathbb{Z}^+$, write down the value of $\sum_{k=1}^{m} \cos\left(\frac{2k\pi}{2m+1}\right)$. [1]

End of question paper

MATHEMATICS

Analysis and approaches

Higher Level | Set A | Paper 3

Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

1. [Maximum mark: 25]

$C:\ x^2+y^2=25$

(a) (i) $2x+2y\dfrac{\mathrm{d}y}{\mathrm{d}x}=0\Rightarrow\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{x}{y}$. **M1 A1**

(ii) $m_T=\left.\dfrac{\mathrm{d}y}{\mathrm{d}x}\right|_{(3,4)}=-\dfrac{3}{4}$ **A1**

$y-4=-\dfrac{3}{4}(x-3)$ **A1**

[4 marks]

$\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{x}{y}$, with $y=4$ when $x=3$.

(b) (i) separable; **YES**. We can separate variables: $y\mathrm{d}y=-x\mathrm{d}x$ **A1**

(ii) homogeneous; **YES,** We can express $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of $v=\dfrac{y}{x}$, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{1}{v}$ **A1**

(iii) linear of first order. **NO**. It does not have the form $\dfrac{\mathrm{d}y}{\mathrm{d}x}+P(x)y=Q(x)$ **A1**

[3 marks]

(c) $y\mathrm{d}y=-x\mathrm{d}x\Rightarrow\int y\mathrm{d}y=-\int x\mathrm{d}x$

$\dfrac{y^2}{2}=-\dfrac{x^2}{2}+c$ **A1**

At (3,4): $\dfrac{4^2}{2}=-\dfrac{3^2}{2}+c\Rightarrow c=\dfrac{25}{2}$ **A1**

Hence, $\dfrac{y^2}{2}=-\dfrac{x^2}{2}+\dfrac{25}{2}\Rightarrow x^2+y^2=25$ **M1**

[3 marks]

(d) (i) $x_0=3,\ y_0=4,\quad x_{n+1}=x_n+0.1,\quad y_{n+1}=y_n-0.1\dfrac{x_n}{y_n}$ **M1**

$y_{10}=3.0324...$ **A1**

(ii) when $x=4$, $y=3$ **A1**

Overestimation since the curve is concave down in the region of the point. **R1**

[4 marks]

(e) (i) $\frac{dy}{dx} = -\frac{x}{y} \Rightarrow \frac{d^2y}{dx^2} = -\frac{y - x\frac{dy}{dx}}{y^2} = -\frac{y + x\frac{x}{y}}{y^2}$ **M1 A1 A1**

Hence, $\frac{d^2y}{dx^2} = -\frac{x^2 + y^2}{y^3}$.

(ii) $f(0) = 5$, $f'(0) = 0$, $f''(0) = -\frac{5^2 - 0}{5^3} = -\frac{1}{5}$ **A1 A1**

Maclaurin: $y = 5 - \frac{1}{10}x^2 + \ldots$ **A1**

(iii) For $x = 4$, $y \cong 5 - \frac{16}{10} = 3.4$ **A1**

[7 marks]

(f) $x^2 + y^2 = 25 \Rightarrow 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow y\frac{dy}{dt} = -x\frac{dx}{dt}$ **M1 A1**

At point P

$\Rightarrow 4\frac{dy}{dt} = -3 \times 8 \Rightarrow \frac{dy}{dt} = -6$ mm per second. **M1 A1**

[4 marks]

2. [Maximum mark: 30]

$z^n = 1$, where $n \geq 2$.

(a) $z_k{}^n = \text{cis}\left(n\dfrac{2k\pi}{n}\right) = \text{cis}\left(2k\pi\right) = 1$. **A1**

[1 mark]

(b) $P_n = -1$ when n is even, $P_n = 1$, when n is odd.

OR otherwise, $P_n = (-1)^{n+1}$ **A1**

[1 mark]

(c) (i) $z_{-1} = \text{cis}\dfrac{-2\pi}{3} = -\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}\text{i}$. $z_0 = 1$, $z_1 = \text{cis}\dfrac{2\pi}{3} = -\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\text{i}$ **A1 A1**

(ii) $P_3 = \left(-\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}\text{i}\right) \times 1 \times \left(-\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\text{i}\right) = \dfrac{1}{4} + \dfrac{3}{4} = 1$. **A2**

[4 marks]

(d) (i) $z_{-1} = -\text{i}$, $z_0 = 1$, $z_1 = \text{i}$, $z_2 = -1$. **A1 A1**

(ii) $P_4 = -\text{i} \times 1 \times \text{i} \times (-1) = \text{i}^2 = -1$. **A1**

[3 marks]

(e) $(z - z_k)(z - z_{-k}) = \left(z - \cos\dfrac{2k\pi}{n} - \text{i}\sin\dfrac{2k\pi}{n}\right)\left(z - \cos\dfrac{2k\pi}{n} + \text{i}\sin\dfrac{2k\pi}{n}\right)$ **A1**

$= \left(z - \cos\dfrac{2k\pi}{n}\right)^2 - \left(\text{i}\sin\dfrac{2k\pi}{n}\right)^2 = z^2 - 2z\cos\dfrac{2k\pi}{n} + \cos^2\dfrac{2k\pi}{n} + \sin^2\dfrac{2k\pi}{n}$ **A1 A1**

$= z^2 - 2z\cos\dfrac{2k\pi}{n} + 1$

[3 marks]

(f) (i) $z_{-2} = \text{cis}\dfrac{-4\pi}{6} = \text{cis}\dfrac{-2\pi}{3}$. $z_{-1} = \text{cis}\dfrac{-2\pi}{6} = \text{cis}\dfrac{-\pi}{3}$, $z_0 = \text{cis}0$,

$z_1 = \text{cis}\dfrac{2\pi}{6} = \text{cis}\dfrac{\pi}{3}$, $z_2 = \text{cis}\dfrac{4\pi}{6} = \text{cis}\dfrac{2\pi}{3}$, $z_3 = \text{cis}\dfrac{6\pi}{6} = \text{cis}\pi$ **A1 A1 A1**

(ii) $z_3 = -1$. **A1**

(iii) $z^6-1=(z-1)(z+1)\left(z^2-2z\cos\frac{\pi}{3}+1\right)\left(z^2-2z\cos\frac{2\pi}{3}+1\right)$ **A1**

$=(z-1)(z+1)\left(z^2-z+1\right)\left(z^2+z+1\right)$ **A1**

[6 marks]

(g) (i) $z_{-2}=\text{cis}\frac{-4\pi}{5}$, $z_{-1}=\text{cis}\frac{-2\pi}{5}$, $z_0=\text{cis}0$, $z_1=\text{cis}\frac{2\pi}{5}$, $z_2=\text{cis}\frac{4\pi}{5}$ **A1 A1**

(ii) $z^5-1=(z-1)\left(z^2-2z\cos\frac{2\pi}{5}+1\right)\left(z^2-2z\cos\frac{4\pi}{5}+1\right)$, **A1 A1**

[4 marks]

(h) $z^7-1\ (z-1)\left(z^2-2z\cos\frac{2\pi}{7}+1\right)\left(z^2-2z\cos\frac{4\pi}{7}+1\right)\left(z^2-2z\cos\frac{6\pi}{7}+1\right)$ **A2**

[2 marks]

(i) The real part of $S_5=0$ gives **M1**

$\cos\frac{-4\pi}{5}+\cos\frac{-2\pi}{5}+1+\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}=0$ **A1**

$\Rightarrow\cos\frac{4\pi}{5}+\cos\frac{2\pi}{5}+1+\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}=0$ **A1**

$\Rightarrow 2\cos\frac{4\pi}{5}+2\cos\frac{2\pi}{5}+1=0$ **A1**

$\Rightarrow\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}=-\frac{1}{2}$

[4 marks]

(j) (I) $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}=-\frac{1}{2}$ **A1**

(II) $\cos\frac{2\pi}{2m+1}+\cos\frac{4\pi}{2m+1}+\cos\frac{6\pi}{2m+1}+\cdots+\cos\frac{2m\pi}{2m+1}=-\frac{1}{2}$. **A1**

[2 marks]